ECON 310-01 Term Paper

Fall 2019

The Effect of Foreign Direct Investment on Investment in Canada.

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Introduction:

Foreign Direct Investment (FDI) is a pivotal element in the global economic landscape, with its effects extending beyond national borders. This paper delves into the intricate relationship between FDI and investment in Canada, focusing on the manufacturing sector. The study employs econometric analysis to unravel the direct impact of FDI on business gross fixed capital formation, considering variables such as Gross National Expenditure (Yt), Full Capacity Output (Ut), and Foreign Direct Investment (FDI).

The econometric equation considered in this paper is:

FUNCTIONAL FORM

$$I_t = \hat{\beta}_0 + \hat{\beta}_1 Y_t - \hat{\beta}_2 U_t + \hat{\beta}_3 FDI_t$$

Where:

Business Gross Capital Fixed Formation (I)= f (Gross National Expenditure (Y_t), Full Capacity Output -Gross National Expenditure(U_t), Foreign Direct Investment (FDI_t)

1. Y_t = Gross National Expenditure

   I expect the sign to to be positive because growing consumption leads to a higher demand of goods and causes firms to invest in capital to satisfy those demands.

2. U_t = Full Capacity Output -Gross National Expenditure

   I expect the sign to be negative because if the country is already at full capacity, the trickledown effect of foreign investment will increase domestic expenditure which will lead to demand-push inflation. Likewise, if the firms

that received foreign Investment work with a lot of imported foreign capital, the added investment will cause higher imports of capital and cause a negative impact on the local capital formation.

3. FDI_t = Foreign Direct Investment

I expect the sign to be positive because the foreign funds used for investment can increase the demand of locally produced domestic capital goods or introduce new tech or cheaper raw materials to produce more capital goods.

Importance:

This topic is important because it is widely considered that Foreign Direct Investment plays an important role in deciding the health of an economy. It is believed that it increases the money supply of the economy, increases job supplies and also standard of living. Considering these wide effect of foreign direct investment to the host country, this paper considers in investigating the direct effects of foreign direct investment in the host country, Canada. This paper spefically investigates on the manufaturing industries of Canda.

Relevance:

As the world became more globalized, so did its economy. This development of modern globalized economy has made a significant positive impact on the living standard of people around the globe. Foreign Direct Investment seems to be a kep player for it. Especially, in terms of development in third world countries, FDI has made a huge positive impact. Countries such as India, China, South Korea has gained the status of a developed countries due to the influx of foreign investments.

Direct and Indirect Impact:

The host counrty, Canada is directly effected by these factors. The socio-economic conditions of the people residing in the host country, local business and foreign business will be directly effected. Local businesses that could potentially used their goods and their neighbouring nations including their trade partners will be indirectly effected.

Motivation

Foreign Direct Investment has always been a topic of my interest. Residing from Nepal, Foreign Direct Investment has lately been having a lot of economic effect in my country. It has brought significant postivie development in my country which I would want to further understand about it.

THEORETICAL Consideration:

a. **Proposed Econometric Specification**

$I_t = f(Y_t ; U_t ; FDI_t)$

Where,

I = Business Gross Fixed Capital Formation ($)

Yt = Gross National Expenditure ($)

Ut = YFt – Yt

YFt = Full capacity output ($)

FDI = Foreign Direct Investment ($)

- **All Variables, Source and Expected Signs**

- **I** = Business Gross Capital Fixed Formation also known as net investment is the net amount of fixed capital accumulation. It measures the increase in the capital stock less the disposal of fixed assets. It is the dependent variable of the model. The source of the data were from the a *Canadian National Accounts*.
- $\mathbf{Y_t}$ (+) = Gross National Expenditure

  Is combined amount if all the expenditures to include those which are public and private. It differs from the GDP in that expenditures on exports are not included. I expect the sign to to be positive because growing consumption leads to a higher demand of goods and causes firms to invest in capital to satisfy those demands. The source of the data were from the Canadian Balance of International Payments and International Investment Position.
- $\mathbf{U_t}$ (-) = Full Capacity Output -Gross National Expenditure, This refers as the accele

  I expect the sign to be negative because if the country is already at full capacity, the trickledown effect of foreign investment will increase domestic expenditure which will lead to demand-push inflation. Likewise, if the firms that received foreign Investment work with a lot of imported foreign capital, the added investment will cause higher imports of capital and cause a negative impact on the local capital formation.

- $\mathbf{FDI_t}$ (+) = Foreign Direct Investment

  I expect the sign to be positive because the foreign funds used for investment can increase the demand of locally produced domestic capital goods or introduce new tech or cheaper raw materials to produce more capital goods.

Descriptive Statistics

Date: 11/17/19 Time: 21:11
Sample: 1981 2018

	I	YT	UT	FDI
Mean	201.8816	900.4490	-476.4514	25.32500
Median	134.9685	646.2120	-539.1710	10.58000
Maximum	447.9230	1872.970	33.02900	120.4200
Minimum	71.01160	301.3270	-792.8020	0.090000
Std. Dev.	127.5662	521.6136	224.2328	28.09576
Skewness	0.768787	0.681328	0.935326	1.451722
Kurtosis	2.031777	1.965631	2.999188	4.821195
Jarque-Bera	4.952387	4.390128	5.249007	17.62011
Probability	0.084063	0.111351	0.072476	0.000149
Sum	7267.738	32416.16	-17152.25	911.7000
Sum Sq. Dev.	569559.3	9522828.	1759812.	27628.01
Observations	36	36	36	36

Index: (in Billions USD)
I = Business Gross Fixed Capital Formation
Yt = Gross National Expenditure
Ut = YFt-Yt; where YFt = Full Capacity Output
FDI = Foreign Direct Investment

Multicollinearity Testing

1. **Correlation Coefficient**

	YT	UT	FDI
YT	1.000000	0.835549	0.723889
UT	0.835549	1.000000	0.515652
FDI	0.723889	0.515652	1.000000

2. **Variance Inflation Factors**

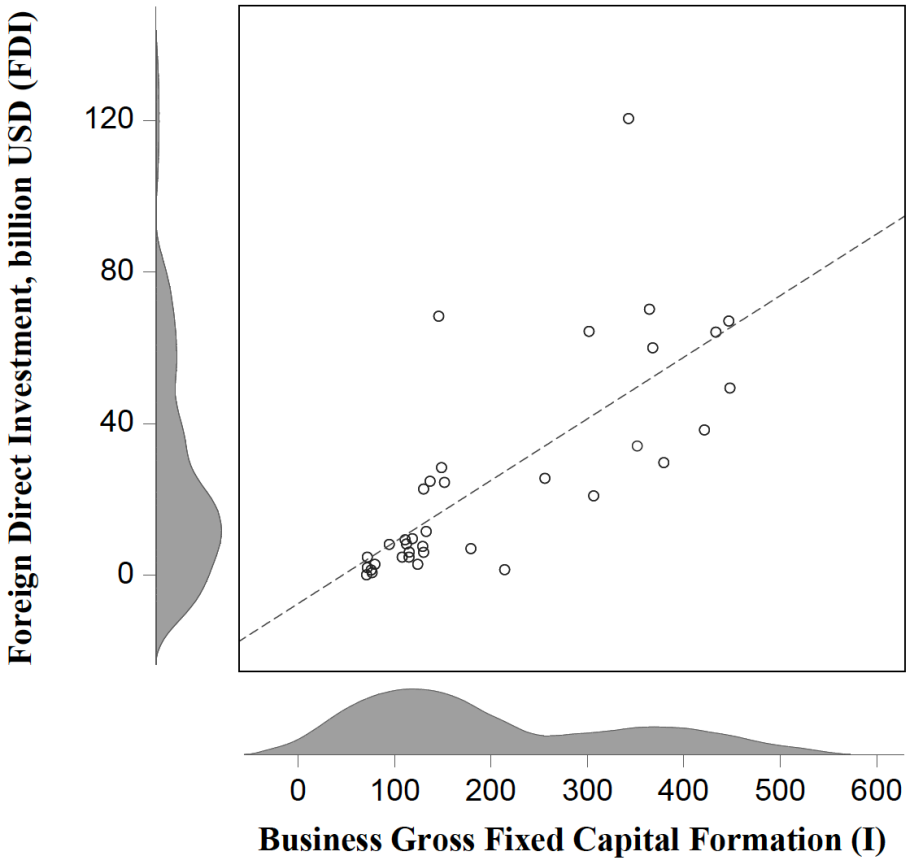
Variance Inflation Factors
Date: 11/17/19 Time: 21:00
Sample: 1981 2018
Included observations: 36

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	89.53445	51.92311	NA
YT	3.53E-05	21.98761	5.408778
UT	0.000124	19.79277	3.506994
FDI	0.004997	4.082696	2.224051

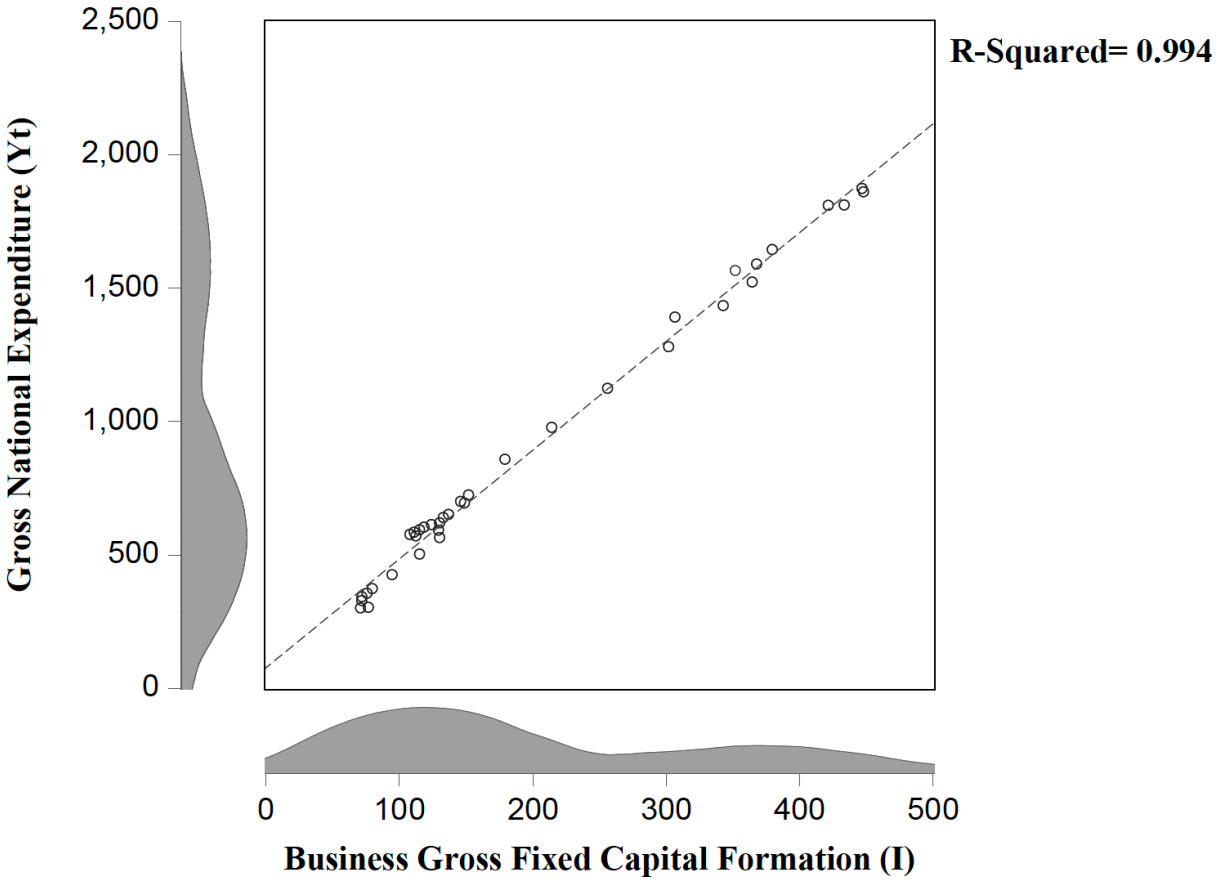
Evaluation

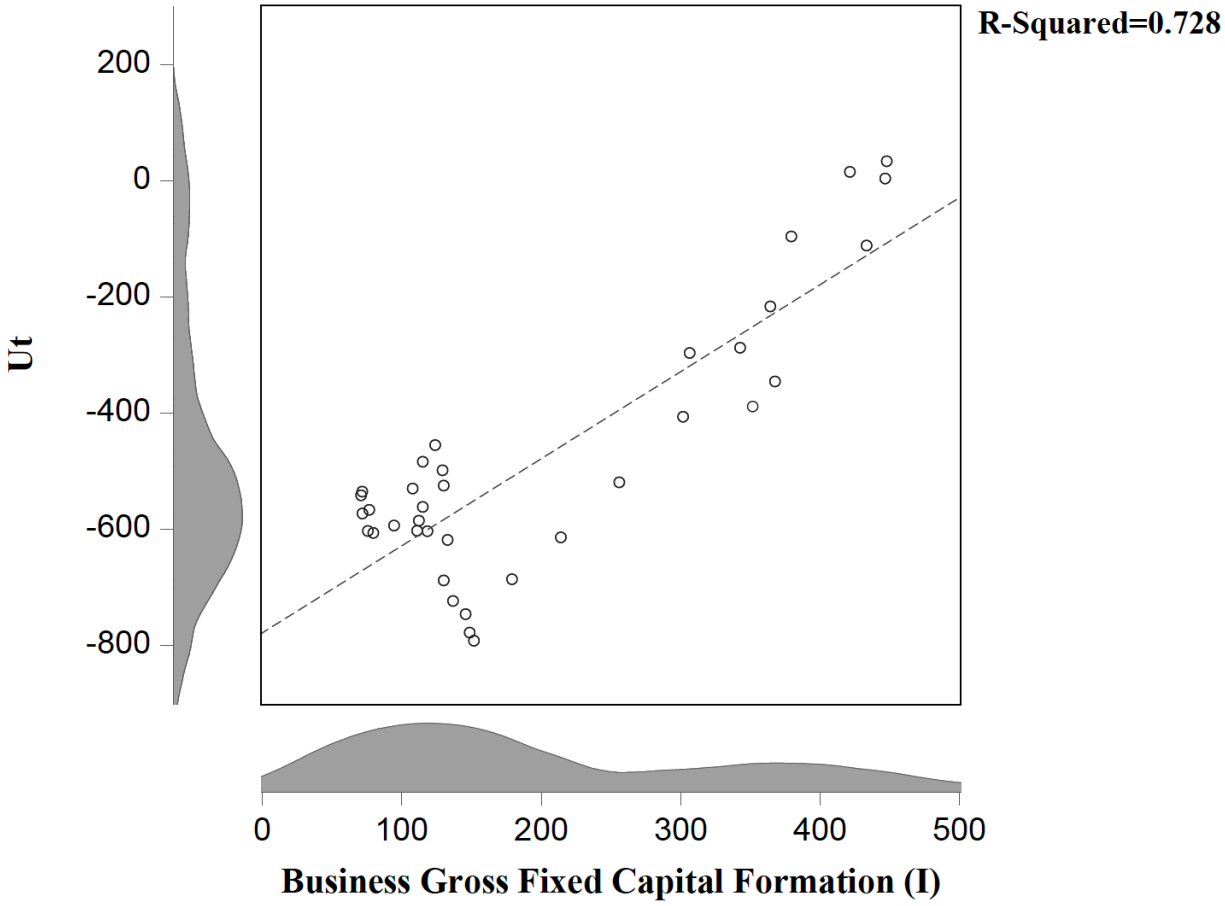
According to the correlation coefficient measure the independent variables yt and ut seemes to show multicollinearity problem. Where as through variance inflation factors we can see that the centered VIF for Yt is grater than 5 which suggest that there is critical level of multicollinearity where the coefficients are poorly estimated, and the p-values are questionable. As for the independent variables Ut and FDI, the VIFs is between 1 and 5 which suggest that there is a moderate correlation, but it is not severe enough to warrant corrective measures.
Correction: This problem can be addressed by adding more independent variables which goes along with our economic theorey. Furthermore, I will update the insuffucient datas in the variable.

Scatterplot Diagrams



R-squared= 0.544


R-Squared= 0.994
Gross National Expenditure (Yt)
2,500
2,000
1,500
1,000
500
0
0
100
200
300
400
500
Business Gross Fixed Capital Formation (I)


R-Squared=0.728
200
0
-200
-400
-600
-800
Ut
0
100
200
300
400
500
Business Gross Fixed Capital Formation (I)

Dependent Variable: I
Method: Least Squares
Date: 11/17/19 Time: 21:04
Sample (adjusted): 1981 2016
Included observations: 36 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	20.92277	9.462265	2.211180	0.0343
YT	0.219011	0.005938	36.88356	0.0000
UT	0.045622	0.011122	4.101823	0.0003
FDI	0.216688	0.070691	3.065286	0.0044

R-squared	0.996512	Mean dependent var	201.8816
Adjusted R-squared	0.996185	S.D. dependent var	127.5662
S.E. of regression	7.878907	Akaike info criterion	7.070695
Sum squared resid	1986.470	Schwarz criterion	7.246641
Log likelihood	-123.2725	Hannan-Quinn criter.	7.132105
F-statistic	3047.673	Durbin-Watson stat	0.775696
Prob(F-statistic)	0.000000		

Theoritical Evaluation
All the variables agree with our theoritical expectation. I expected all the variables to have positive effect on the dependent variables and, hence expected to have positive signs which it does.

Statistically Significance
All the coefficient seems to be statistically significant as they maintain significant level of approximately 0.05 and above.

Impact on Independent Variables:

- Yt, Gross National Expenditure.
  The equation suggest that for each $1.00 change in the Gross National Expenditure (Yt) then there will be an increase of $0.219 in the Business Gross Fixed Capital Formation.
- Ut, Yft-Yt
  The equation suggest that for $1.00 change in the Ut then there will be an increase of $0.045 USD in the Business Gross Fixed Capital Formation.
- FDI, Foreign Direct Investment
  The equation suggest that for $1.00 change in the Foreign Direct Investment (FDI) then there will be an increase of $0.216 USD in the Business Gross Fixed Capital Formation.

Overall Fit and Significance:
The adjusted R-Squared is 0.996 which states that the independent variables involved explains 99.6% of the variation in the dependent variable. The probability of the F-statistics is less than 0.1 which suggest that the R-Squared is statistically significant.

Durbin Wtason Test:
Durbin Watson Statistics: 0.775
N= 36
K=3
Appropriate Values:
du=1.65, dl=1.30, 4-du = 2.35 and 4-di = 2.70
The Durbin Watson test suggest that there is an evidence of positive auto-correlation and hence he reject out Ho hypothesis. This is because our Durbin Watson Statistics for the equation is less than du.

Granger Casuality Test:

Pairwise Granger Causality Tests
Date: 12/01/19 Time: 15:40
Sample: 1981 2018
Lags: 1

Null Hypothesis:	Obs	F-Statistic	Prob.
YT does not Granger Cause I	35	4.34164	0.0453
I does not Granger Cause YT		2.79059	0.1046

- The result of GCT tells us that the YT granger causes I. Therefore, we reject null hypothesis.
- Whereas the I does not granger cause YT. Therefore, we accept the null hypothesis.
- Therefore, it appears that the granger causality runs one way from YT to I.

Pairwise Granger Causality Tests
Date: 12/01/19 Time: 15:38
Sample: 1981 2018
Lags: 2

Null Hypothesis:	Obs	F-Statistic	Prob.
YT does not Granger Cause I	34	5.71224	0.0081
I does not Granger Cause YT		3.65169	0.0385

- The result of GCT tells us that the YT granger causes I. Therefore, we reject null hypothesis.
- Similarly, the I does granger cause YT. Therefore, we reject the null hypothesis.
- Therefore, it appears that the granger causality runs both way from YT to I and vice versa.

Pairwise Granger Causality Tests
Date: 12/01/19 Time: 15:39
Sample: 1981 2018
Lags: 1

Null Hypothesis:	Obs	F-Statistic	Prob.
UT does not Granger Cause I	35	1.78355	0.1911
I does not Granger Cause UT		0.75759	0.3906

- The result of GCT tells us that the UT does not granger causes I. Therefore, we accept null hypothesis.
- Similarly, the I does not granger cause UT. Therefore, we accept the null hypothesis.
- Therefore, it appears that the granger causality runs both way from UT to I and vice versa.

Pairwise Granger Causality Tests
Date: 12/01/19 Time: 15:40
Sample: 1981 2018
Lags: 2

Null Hypothesis:	Obs	F-Statistic	Prob.
UT does not Granger Cause I	34	5.11882	0.0125
I does not Granger Cause UT		6.50581	0.0046

- The result of GCT tells us that the UT granger causes I. Therefore, we reject null hypothesis.
- Similarly, the I does granger cause UT. Therefore, we reject the null hypothesis.
- Therefore, it appears that the granger causality runs both way from UT to I and vice versa.

Pairwise Granger Causality Tests
Date: 12/01/19 Time: 15:38
Sample: 1981 2018
Lags: 1

Null Hypothesis:	Obs	F-Statistic	Prob.
FDI does not Granger Cause I	35	1.43014	0.2405
I does not Granger Cause FDI		5.26207	0.0285

- The result of GCT tells us that the FDI does not granger causes I. Therefore, we fail to reject null hypothesis.

- Whereas, the I does granger cause FDI. Therefore, we reject the null hypothesis.
- Therefore, it appears that the granger causality runs one way from I to FDI.

Pairwise Granger Causality Tests
Date: 12/01/19 Time: 15:34
Sample: 1981 2018
Lags: 2

Null Hypothesis:	Obs	F-Statistic	Prob.
FDI does not Granger Cause I	34	1.36361	0.2716
I does not Granger Cause FDI		5.40679	0.0101

- The result of GCT tells us that the FDI does not granger causes I. Therefore, we fail to reject null hypothesis.
- Whereas, the I does granger cause FDI. Therefore, we reject the null hypothesis.
- Therefore, it appears that the granger causality runs one way from I to FDI.

Revision of GCT Test.
I will be revising the GCT test because enough evidence have been provided that some of the independent variables does granger cause the dependent variables.

Omitted Variable Test
Omitted Variable Test
Null hypothesis: FDI
Equation: EQ02_OVT_FDI
Specification: I C YT UT
Omitted Variables: FDI is not significant

	Value	df	Probability
t-statistic	3.065286	32	0.0044
F-statistic	9.395979	(1, 32)	0.0044
Likelihood ratio	9.268122	1	0.0023

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	583.2759	1	583.2759
Restricted SSR	2569.746	33	77.87108
Unrestricted SSR	1986.470	32	62.07718

LR test summary:

	Value	df
Restricted LogL	-127.9066	33

Unrestricted LogL	-123.2725	32

Unrestricted Test Equation:
Dependent Variable: I
Method: Least Squares
Date: 11/18/19 Time: 14:44
Sample: 1981 2016
Included observations: 36

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	20.92277	9.462265	2.211180	0.0343
YT	0.219011	0.005938	36.88356	0.0000
UT	0.045622	0.011122	4.101823	0.0003
FDI	0.216688	0.070691	3.065286	0.0044

R-squared	0.996512	Mean dependent var	201.8816
Adjusted R-squared	0.996185	S.D. dependent var	127.5662
S.E. of regression	7.878907	Akaike info criterion	7.070695
Sum squared resid	1986.470	Schwarz criterion	7.246641
Log likelihood	-123.2725	Hannan-Quinn criter.	7.132105
F-statistic	3047.673	Durbin-Watson stat	0.775696
Prob(F-statistic)	0.000000		

Omitted Variable Test
Null hypothesis: UT
Equation: UNTITLED
Specification: I C YT FDI
Omitted Variables: UT is not significant

	Value	df	Probability
t-statistic	4.101823	32	0.0003
F-statistic	16.82495	(1, 32)	0.0003
Likelihood ratio	15.21020	1	0.0001

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	1044.445	1	1044.445
Restricted SSR	3030.915	33	91.84591
Unrestricted SSR	1986.470	32	62.07718

LR test summary:

	Value	df
Restricted LogL	-130.8776	33
Unrestricted LogL	-123.2725	32

Unrestricted Test Equation:
Dependent Variable: I
Method: Least Squares
Date: 11/18/19 Time: 14:42
Sample: 1981 2016
Included observations: 36

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	20.92277	9.462265	2.211180	0.0343
YT	0.219011	0.005938	36.88356	0.0000
FDI	0.216688	0.070691	3.065286	0.0044
UT	0.045622	0.011122	4.101823	0.0003

R-squared	0.996512	Mean dependent var	201.8816
Adjusted R-squared	0.996185	S.D. dependent var	127.5662
S.E. of regression	7.878907	Akaike info criterion	7.070695
Sum squared resid	1986.470	Schwarz criterion	7.246641
Log likelihood	-123.2725	Hannan-Quinn criter.	7.132105
F-statistic	3047.673	Durbin-Watson stat	0.775696
Prob(F-statistic)	0.000000		

Omitted Variable Test
Null hypothesis: YT
Equation: UNTITLED
Specification: I C UT FDI
Omitted Variables: YT is not significant

	Value	df	Probability
t-statistic	36.88356	32	0.0000
F-statistic	1360.397	(1, 32)	0.0000
Likelihood ratio	135.8297	1	0.0000

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	84449.62	1	84449.62
Restricted SSR	86436.09	33	2619.275
Unrestricted SSR	1986.470	32	62.07718

LR test summary:

	Value	df
Restricted LogL	-191.1873	33
Unrestricted LogL	-123.2725	32

Unrestricted Test Equation:
Dependent Variable: I
Method: Least Squares
Date: 11/18/19 Time: 14:43
Sample: 1981 2016
Included observations: 36

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	20.92277	9.462265	2.211180	0.0343
UT	0.045622	0.011122	4.101823	0.0003
FDI	0.216688	0.070691	3.065286	0.0044
YT	0.219011	0.005938	36.88356	0.0000

R-squared	0.996512	Mean dependent var	201.8816
Adjusted R-squared	0.996185	S.D. dependent var	127.5662
S.E. of regression	7.878907	Akaike info criterion	7.070695
Sum squared resid	1986.470	Schwarz criterion	7.246641
Log likelihood	-123.2725	Hannan-Quinn criter.	7.132105
F-statistic	3047.673	Durbin-Watson stat	0.775696
Prob(F-statistic)	0.000000		

Evidence of Significance:
If the F-test p-value is less than 0.1 than it is significant variable. All the variables seems to be significant as the evidence suggests.

Place the EViews results of the Chow Test(s) HERE.

Chow Breakpoint Test: 2008
Null Hypothesis: No breaks at specified breakpoints
Varying regressors: All equation variables
Equation Sample: 1981 2016

F-statistic	0.915543	Prob. F(4,28)	0.4686
Log likelihood ratio	4.425054	Prob. Chi-Square(4)	0.3515
Wald Statistic	3.662173	Prob. Chi-Square(4)	0.4537

1. **What is the reasoning or logic for selecting the observation number (for cross section data) or breakpoint date (for time series data)?**
   The reason I chose 2008 as a breakpoint date because of the housing crisis which negatively affected the foreign direct investment (FDI) in Canada. The FDI in Canada decreased from 120.42 billion USD in 2007 to 70.12 Billion USD in 2008.

2. **Interpret the test: did the Chow test confirm or fail to confirm your hypothesis?**
   Through the test we can confirm our hypothesis that there was a specified breakpoint in 2008. We can see that the Prob.F (4, 28) is 0.4686 which is more than 0.10. Thus, no eveidence of a break in 2008.

Dependent Variable: I

Method: ARMA Maximum Likelihood (OPG - BHHH)

Date: 12/01/19 Time: 15:28

Sample: 1981 2016

Included observations: 36

Convergence achieved after 12 iterations

Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	23.63189	14.04215	1.682925	0.1028
YT	0.221004	0.007522	29.38219	0.0000
UT	0.049600	0.018391	2.696932	0.0114
FDI	0.136900	0.057726	2.371535	0.0243
AR(1)	0.660511	0.142920	4.621547	0.0001
SIGMASQ	34.04448	9.052525	3.760772	0.0007

R-squared	0.997848	Mean dependent var	201.8816
Adjusted R-squared	0.997490	S.D. dependent var	127.5662
S.E. of regression	6.391665	Akaike info criterion	6.714800
Sum squared resid	1225.601	Schwarz criterion	6.978720
Log likelihood	-114.8664	Hannan-Quinn criter.	6.806915
F-statistic	2782.309	Durbin-Watson stat	2.048368
Prob(F-statistic)	0.000000		

Inverted AR Roots	.66

- **Changes to initial model.**

As a change from the initial model, I corrected the serial correlation present in the model by adding an AR (1) term. After adding **AR (1)** in the equation we find that the adjusted r-square increased to 0.997 and F-stat decreased to 3047.673. Whereas, the DW stat decreased to 2.048 which is lower than 4-du = 2.35 and higher than du= 1.65. Therefore, we accept the null hypothesis and there is no negative FOSC. This means that the inclusion of AR (1) has improved our equation.

- **Schwarz and Akaike Information Criteria.**

We prefer the Schwarz criterion and Akaike info criterion to be low. Hence, we will pick the mode with lower values. Both of these criteria are used to estimate the out-of-sample forecast error variance, like the Standard Error of the regression, and penalize degrees of freedom.
The Schwarz criterion has decreased in the model to 6.97 after the changes. Similarly, the Akaike info criterion has decreased to 6.71. Therefore, we can say that the Schwarz criterion and Akaike info criterion has helped in our model development and final specification.

Re-run of appropriate diagnostic test.

Variance Inflation Factors
Date: 12/01/19 Time: 17:14
Sample: 1981 2018
Included observations: 36

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	197.1821	32.63788	NA
YT	5.66E-05	8.568927	2.593310
UT	0.000338	15.40098	2.597225
FDI	0.003332	1.539959	1.539827
AR(1)	0.020426	1.097468	1.090799
SIGMASQ	81.94821	1.477314	1.421783

By adding AR (1) in the model, we have corrected the multicollinearity problem. Previously, our model showed moderate to critical level of multicollinearity between the variables which need

corrective measures. Now, all the variables have VIFs between 1 and 5 which suggest that there is just a moderate correlation and does not require corrective measures.

Discernable Deficiencies.

I do not see any major discernable deficiencies in my model. Before adding the AR (1) term, my model suffered from correlation issue. However, the addition of the AR (1) has brought significant positive changes to my model.

Employment of the Model.

I believe that the government, domestic manufacturing companies and foreign investing entities will generate interest in the model. The paper studies the direct effects of foreign direct investment on the business gross fixed capital formation.

According to my model, foreign direct investment, gross national expenditure and full capacity output of the nation is the major factor for the investment in the country thereby its ability to produce. This can help the country to maintain a positive balance of payments. Hence, based on my model, the government of the host country could closely evaluate the effect of foreign direct investment as these factors can have a huge implication in the nation's economy. Likewise, foreign investing entities could look at this model in order to examine the profitability of their investment. Furthermore, the domestic manufacturing companies could use this as a study to improve their ability to attract foreign investments.

Potential Risks.

One issue that I find in my model while trying to formulate a conclusion is that currently, the model measures the direct impact of foreign direct investment in Canada. However, it ignores the indirect effect of the foreign direct investment in Canada which can have negative or positive outcome. Hence, to have much more useful use of this model, one needs to include studies for indirect spillover effect of foreign direct investment in the country.

Conclusion.

The study sole purpose was to measure the direct impact of foreign direct investment in Canada. This was done by considering ways in which foreign direct investment affects total investment in Canada. In our model, the coefficient of direct investment is the focus of our interest. If the coefficient is significantly greater than one, we come to a conclusion that the domestic investment has increased as a result of the rise in foreign investment, i.e., foreign investment has given rise to complementary effects. Whereas if the coefficient is significantly less than one, domestic investment has decreased as a result of the rise in foreign investment, i.e., foreign investment has substituted for some domestic investment.

As expected through our economic theory, the variable FDI holds a positive relationship with the investment in the country. Specifically, the study shows that a $1 of foreign direct investment caused an increase of $0.14 of total investment through direct effect. The result of the study suggests that the foreign direct investment is statistically significant, but however it has decreased the domestic investment as a result of rise in foreign investment. This proves that the foreign direct investment does not have a complimentary factor on total investment as it does not stimulate domestic investment.

However, I believe that our model is incomplete to wholly understand the effect of foreign direct investment as it ignores the indirect effect of the foreign direct investment in Canada which can have negative or positive outcome. Hence, to have much more useful use of this model, one needs to include studies for indirect spillover effect of foreign direct investment in the country. Hence, I suggest the policy makers not to simply assume that the foreign direct investment can have positive influence in the economy based on this study, but rather conduct further study on the indirect effect of FDI and compare it with this model. It is critically important to examine the precisely how these foreign funds are used.

Comparison and Contrast.

The data collected from my literature review was for the time period of 1948-1966 while the data I collected was from1981-2018. This data sample resulted in the following estimated equations:

Reference Eqn: -449 + 0.208 Yt - 0.329 Ut + 1.398 FDI

(429.) (0.015) (0.76) (0.375)

$R^2 = 0.9507$ DW=1.65

Estimated Eqn: 23.63 + 0.221 Yt + 0.05 Ut + 0.137 FDI + 0.66 AR(1)

(1.68) (29.38) (2.69) (2.37) (4.62)

$R^2 = 0.998$ DW=2.05

Comparing the above equations, I found that my equation has slightly higher r-bar squared. In order words, the fit of my model states that my set of independent variables can explain about 99.8% of the variation. However, in comparison with the reference equation the coefficient of variables my model has decreased. As we solely focused on the coefficient of foreign direct investment in our model, we can see that the coefficient of FDI decreased to 0.137 which lower than our parameter of 1. Hence, in comparison the effects of FDI to total investment has shifted in these tow models. At first, the coefficient of FDI suggested that there is high probability that the FDI produces complementary effect, i.e., it stimulates additional domestic investment. However, in my model, we come to conclusion that the FDI does not stimulate additional domestic investment, it rather decreases.

Broader Political and Social Implications.

It evident through the study that the foreign direct investment and the GNE of the host nation's is a significant contributor in determining the possible effect on the total investment in the country. My study proves that although there might be positive correlation between fdi and the investment

in the country, it does not stimulate domestic investment. Hence, this information can be very useful while implementing policies in host nations. It could be used to implement policies that would protect and incentivize domestic investment while not harming foreign direct investment. This would help to bring complementary shift to the model.

Area of Future Research.

My model only explains the direct effect of Foreign Direct Investment in the economy. Hence, an area of future research that I would like to explore would be to look at the indirect spillover effect of Foreign Direct Investment.